

Sharp and Computable Corrections for Nonuniform Coupon Collection

A positive remainder bound, optimal kernel constants,
and a deterministic algorithm with probability enclosures

Research prepared for Vladimir Reshetnikov

8 October 2026

Abstract

The probability of observing every category in a fixed number of independent samples is easy to approximate after Poissonization, but a numerical approximation is substantially more useful when accompanied by a computable error bound. We refine a recent Taylor-remainder lemma of Hwang, Li, and Zacharovas for the kernel $(1 - x)^m e^{mx}$. A change of variable with nonnegative Taylor coefficients produces an explicit pointwise bound that is smaller throughout the original domain. Its integration yields a positive moment certificate for arbitrary coupon probabilities.

For truncation before degree $2s$, we determine the optimal scalar remainder constant through its second asymptotic term:

$$\mathcal{K}_{m,s} = \frac{m^s}{2^s s!} \left(1 + \frac{s(-2s^2 + 5s + 1)}{9m} + \mathcal{O}_s(m^{-2}) \right).$$

The maximizing argument is uniquely defined for sufficiently large m and equals $2s(s + 1)/(3m) + \mathcal{O}_s(m^{-2})$. Both quantities admit convergent expansions in $1/m$ near infinity at each fixed order. The positive moment certificate is asymptotically attained by a rare-coupon family.

All derivatives and moments needed by the certificate are computed by truncated products in $\mathcal{O}(ns^2)$ arithmetic operations, after an order-dependent setup. An Arb implementation encloses both truncation and rounding errors. We also give explicit bounds uniform over every probability vector in the region $\sum_i e^{-mp_i} \leq L$, recover the sharp scale $(\log n/n)^s$, and provide reproducible numerical checks. The Poisson–Charlier expansion itself is classical; the research contribution is the refined remainder analysis and its constructive, certified use.

Complete proofs and executable verification accompany this manuscript. The precise comparison with prior work is stated in Section 1.2; the asymptotic results keep s fixed.

Contents

1	The problem and the result	2
1.1	What is proved here	2
1.2	Position relative to the literature	2
1.3	Scope of the claims	3
2	Exact identities and notation	3
3	A positive-coefficient remainder bound	4
3.1	The classical Taylor coefficients	4
3.2	A change of variable	5
4	The certified coupon probability	6
4.1	The first useful orders	6
4.2	Comparison of the integrated bounds	7
5	The optimal scalar constant	7
5.1	Its leading term is best possible	7
5.2	A second term and a unique worst argument	8
5.3	A simple finite improvement at order one	10
6	Uniform control for arbitrary probability vectors	10
7	Sharpness within the coupon model	12
7.1	A rare coupon attains the certificate	12
7.2	The uniform rate has the correct order	13
8	Computation in linear work at fixed order	14
8.1	Products replace exponentially many subsets	14
8.2	Rounding belongs in the certificate	14
8.3	What the algorithmic statement means	15
9	Reproducible computations	15
9.1	Validation independent of the approximation	15
9.2	Uniform coupons	16
9.3	Sharpness and unequal probabilities	16
9.4	Running time and numerical environment	17
10	Consequences and practical interpretation	18
10.1	From probabilities to sample-size decisions	18
10.2	Why more moments improve a bound	18
10.3	An all-orders scalar calculation	19
11	Further research questions	19
A	Reproduction and file guide	20
B	A compact checklist of theorem domains	21

1 The problem and the result

Suppose that each independent observation belongs to one of n categories, with known probabilities $p_1, \dots, p_n$. How likely is it that all categories have appeared after m observations? This is the nonuniform coupon collector problem. It is also a finite occupancy question: place m independent balls into n boxes, choosing box i with probability p_i , and ask whether every box is occupied.

Write T for the first time at which every category has appeared, and put $P_m = \mathbb{P}(T \leq m)$. The input assumptions throughout are

$$n \geq 1, \quad p_i \geq 0, \quad \sum_{i=1}^n p_i = 1, \quad m \in \mathbb{N}, \quad m \geq 1. \quad (1)$$

Zero probabilities and the condition $m < n$ are allowed. They make $P_m = 0$, but allowing them in the theorems is useful for checking boundary cases.

If the number of observations is instead Poisson with mean t , the box counts are independent Poisson variables. The resulting coverage probability is the elementary product

$$F(t) = \prod_{i=1}^n (1 - e^{-tp_i}). \quad (2)$$

Replacing the fixed sample size by a Poisson sample size is called *Poissonization*; recovering fixed-sample information is *depoissonization*. A product such as (2) can be evaluated with one pass over the probabilities. The challenge is to retain this computational simplicity while controlling the error at a fixed sample size.

1.1 What is proved here

The central finite result is the interval

$$P_m \in [A_s(m) - R_s(m), A_s(m) + R_s(m)] \cap [0, 1], \quad (3)$$

where A_s is a classical Poisson–Charlier approximation and R_s is an explicit positive expression. It applies to every input in (1). It does not require comparable probabilities, a limit distribution, or a large sample size. Those assumptions enter only when describing how small the interval becomes asymptotically.

There are three complementary conclusions.

1. **A stronger finite bound.** The scalar inequality underlying (3) improves an explicitly identified remainder lemma on its full domain. It leads to a certificate involving moments through order $3s$, all computed without enumerating subsets.
2. **Optimal constants.** The best scalar constant has an explicit two-term asymptotic expansion. A genuine coupon-collector family attains the leading constant in the positive moment certificate.
3. **An implementable calculation.** For each fixed s , the arithmetic work grows linearly with n . The supplied interval-arithmetic program returns a probability enclosure, and separate exact calculations check the implementation on small instances.

1.2 Position relative to the literature

Classical analytic depoissonization already supplies systematic expansions in derivatives of the Poisson transform [4]. High-order approximations for uniform coupon collection are also established [8]. Recent elementary work gives explicit remainders using finite differences [9].

Consequently, the expansion and the existence of fixed-order asymptotic corrections are prior results.

The closest pointwise comparison is Lemma 5.3, equation (46), of the May 2026 preprint of Hwang, Li, and Zacharovas [3]. With the variables renamed as in this paper, it bounds a Taylor remainder by

$$|(1-x)^m e^{mx} - Q_{m,s}(x)| \leq 2s 4^s m^s x^{2s}, \quad 0 \leq x \leq 1, \quad (4)$$

where $Q_{m,s}$ is the Taylor polynomial through degree $2s-1$. Theorem 3.2 replaces the coefficient by a smaller explicit polynomial in x . For $s=1$, their earlier Lemma 2.1 has the better bound mx^2 ; our coefficient is $m(1+x)/2$, which improves that comparison for $0 < x < 1$ as well. We compare with the stated lemmas, rather than claim a general improvement of every result in those papers.

The supplied repository [7] motivated a search for a quantitative theorem with an independently checkable proof and reusable computational artifacts. Its approximate-counting families 113–115 suggested the guiding question: can a combinatorial probability be approximated efficiently while retaining an explicit error guarantee? All mathematical inputs used below are proved in the manuscript or attributed to the established literature; the repository’s larger conjecture-resolution claims are not dependencies of our proofs. The snapshot inspected was commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

1.3 Scope of the claims

The result is a concrete strengthening of an existing quantitative lemma, together with sharp asymptotic analysis of its optimal constant. It does not resolve a longstanding major open conjecture. The literature review identified the overlap above; it cannot establish that every variant of the remainder estimate is absent from all previous publications.

The certificate controls *absolute* error. It can be uninformative early in the collection process, and it does not automatically give small relative error when P_m is extremely small. All asymptotic expansions in the truncation order keep s fixed. The operation count is an arithmetic count with exponential evaluations, not a theorem about bit complexity. These distinctions are part of the mathematical statements, not exceptions to them.

2 Exact identities and notation

For $A \subseteq [n]$, let $p_A = \sum_{i \in A} p_i$. The event that every category in A is missing after m observations has probability $(1-p_A)^m$. Inclusion–exclusion therefore gives

$$P_m = \sum_{A \subseteq [n]} (-1)^{|A|} (1-p_A)^m. \quad (5)$$

Expanding (2) gives the corresponding exponential sum:

$$F(t) = \sum_{A \subseteq [n]} (-1)^{|A|} e^{-tp_A}. \quad (6)$$

These formulas show directly that $F(t) = e^{-t} \sum_{m \geq 0} P_m t^m / m!$ for $t \geq 0$, with the natural convention $P_0 = 0$. They also avoid needing an independence argument for Poisson splitting in the proofs that follow.

Define the unsigned subset moments

$$M_k(t) = \sum_{A \subseteq [n]} p_A^k e^{-tp_A}, \quad k \geq 1. \quad (7)$$

The term $A = \emptyset$ is zero. The moments are nonnegative and satisfy

$$M_{k+1}(t) \leq M_k(t), \quad k \geq 1, \quad (8)$$

because $0 \leq p_A \leq 1$. In contrast,

$$(-1)^k F^{(k)}(t) = \sum_A (-1)^{|A|} p_A^k e^{-tp_A} \quad (9)$$

can have either sign. The unsigned moments preserve a rigorous error bound when the inclusion–exclusion terms cancel.

Remark 2.1 (A larger class). Let μ be a finite signed or complex measure on $[0, 1]$, and define

$$a_m = \int (1-x)^m d\mu(x), \quad f(t) = \int e^{-tx} d\mu(x), \quad M_k^\mu(t) = \int x^k e^{-tx} d|\mu|(x).$$

The same proofs apply with these definitions. For coupon collection, $\mu = \sum_A (-1)^{|A|} \delta_{p_A}$. If distinct subsets have the same total probability, opposite signs may cancel in μ . The explicit M_k in (7) is still a valid positive majorant for M_k^μ and is easier to compute by products.

3 A positive-coefficient remainder bound

3.1 The classical Taylor coefficients

For an integer $m \geq 1$, define

$$E_m(x) = (1-x)^m e^{mx} = \sum_{k \geq 0} c_k(m) x^k, \quad Q_{m,s}(x) = \sum_{k=0}^{2s-1} c_k(m) x^k. \quad (10)$$

The function E_m is entire. Logarithmic differentiation near the origin gives

$$c_0(m) = 1, \quad c_1(m) = 0, \quad kc_k(m) = -m \sum_{j=0}^{k-2} c_j(m) \quad (k \geq 2). \quad (11)$$

Keeping prefix sums evaluates this recurrence in linear arithmetic work in the number of coefficients. For example,

$$c_2 = -\frac{m}{2}, \quad c_3 = -\frac{m}{3}, \quad c_4 = \frac{m^2}{8} - \frac{m}{4}, \quad c_5 = \frac{m^2}{6} - \frac{m}{5}.$$

We suppress the argument m in short formulas.

Lemma 3.1 (Degree and first omitted coefficient). *For fixed k , $c_k(m)$ is a polynomial in m of degree at most $\lfloor k/2 \rfloor$. In particular,*

$$c_{2s}(m) = \frac{(-1)^s m^s}{2^s s!} + \mathcal{O}_s(m^{s-1}). \quad (12)$$

Proof. For $|x| < 1$,

$$E_m(x) = \exp \left(-m \sum_{j \geq 2} \frac{x^j}{j} \right).$$

Every factor contributing a power of m has degree at least two in x . The coefficient of $m^s x^{2s}$ can therefore only come from s copies of $-mx^2/2$, proving both statements. $\square$

3.2 A change of variable

Put

$$d(x) = 1 - (1 - x)e^x = \sum_{k \geq 2} \frac{k-1}{k!} x^k. \quad (13)$$

All coefficients are nonnegative, $d(1) = 1$, and the coefficient of x^2 is $1/2$. It follows that, on the full interval $0 \leq x \leq 1$,

$$0 \leq d(x) \leq \frac{x^2}{2}(1+x) \leq x^2. \quad (14)$$

Indeed, all terms of degree at least three have total coefficient $1/2$ and are bounded by that coefficient times x^3 .

For $1 \leq j < s$, define the rational constants

$$\theta_{j,2s} = 1 - \sum_{k=0}^{2s-1} [x^k]d(x)^j, \quad B_{m,s} = \sum_{j=1}^{s-1} \binom{m}{j} \theta_{j,2s}. \quad (15)$$

The sum defining $B_{m,1}$ is empty and equals zero. Since the coefficients of d^j are nonnegative and sum to $d(1)^j = 1$,

$$0 \leq \theta_{j,2s} \leq 1. \quad (16)$$

We use the convention $\binom{m}{j} = 0$ for $j > m$.

Theorem 3.2 (Pointwise finite refinement). *For all integers $m, s \geq 1$ and all $0 \leq x \leq 1$,*

$$|E_m(x) - Q_{m,s}(x)| \leq x^{2s} \left(B_{m,s} + \frac{\binom{m}{s}}{2^s} (1+x)^s \right). \quad (17)$$

Proof. First assume $m \geq s$. Taylor's theorem for $(1-y)^m$ on $0 \leq y \leq 1$ gives

$$\left| (1-y)^m - \sum_{j=0}^{s-1} (-1)^j \binom{m}{j} y^j \right| \leq \binom{m}{s} y^s. \quad (18)$$

The absolute value of the s th derivative is at most $s! \binom{m}{s}$ on this interval. If $m < s$, the finite sum already equals $(1-y)^m$, and the same inequality holds with a zero right-hand side.

Substitute $y = d(x)$. The identity $1 - d(x) = (1-x)e^x$ gives $(1-d(x))^m = E_m(x)$. Every omitted power $d(x)^j$ with $j \geq s$ starts at degree at least $2s$. Thus the Taylor polynomial through degree $2s-1$ of the finite sum in (18) is exactly $Q_{m,s}$.

For each $1 \leq j < s$, the discarded coefficient tail satisfies

$$0 \leq d(x)^j - \sum_{k=0}^{2s-1} ([x^k]d(x)^j) x^k \leq \theta_{j,2s} x^{2s}.$$

This uses $x^k \leq x^{2s}$ for $k \geq 2s$ and the nonnegative coefficients. Adding the finite Taylor-tail errors and the binomial remainder gives

$$|E_m(x) - Q_{m,s}(x)| \leq B_{m,s} x^{2s} + \binom{m}{s} d(x)^s.$$

Now apply (14). Every step holds at $x = 1$ as well as in the interior, so there is no endpoint exception. $\square$

Corollary 3.3 (Direct comparison with the prior lemma). *The coefficient in (17) satisfies*

$$B_{m,s} + \frac{\binom{m}{s}}{2^s} (1+x)^s \leq \sum_{j=1}^s \binom{m}{j} \leq m^s \sum_{j=1}^s \frac{1}{j!} < (e-1)m^s. \quad (19)$$

In particular, (17) is a pointwise refinement of (4), with a strictly smaller upper-bound function for $x > 0$. For $s = 1$ it is

$$|E_m(x) - 1| \leq \frac{m}{2} x^2 (1+x). \quad (20)$$

Proof. Use (16) and $(1+x)^s \leq 2^s$. Then use $\binom{m}{j} \leq m^j/j! \leq m^s/j!$ for $j \leq s$ and $m \geq 1$. $\square$

4 The certified coupon probability

Define the approximant and its error radius by

$$A_s(m) = \sum_{k=0}^{2s-1} (-1)^k c_k(m) F^{(k)}(m), \quad (21)$$

and

$$R_s(m) = B_{m,s} M_{2s}(m) + \frac{\binom{m}{s}}{2^s} \sum_{\ell=0}^s \binom{s}{\ell} M_{2s+\ell}(m). \quad (22)$$

Every contribution to (22) is nonnegative.

Theorem 4.1 (Finite probability enclosure). *Under (1), for every integer $s \geq 1$,*

$$|P_m - A_s(m)| \leq R_s(m). \quad (23)$$

The same assertion holds in Remark 2.1, with total-variation moments in place of the unsigned subset moments.

Proof. Multiply (17) by e^{-mx} and set $x = p_A$. Subtract the signed sums in (5) and (9). The triangle inequality bounds the difference by

$$\sum_A e^{-mp_A} p_A^{2s} \left(B_{m,s} + \frac{\binom{m}{s}}{2^s} (1+p_A)^s \right).$$

Expanding $(1+p_A)^s$ gives precisely (22). Integration against $|\mu|$ proves the more general version. $\square$

4.1 The first useful orders

All derivatives and moments in the following formulas are evaluated at $t = m$. At order one,

$$A_1 = F, \quad R_1 = \frac{m}{2} (M_2 + M_3). \quad (24)$$

At order two,

$$A_2 = F - \frac{m}{2} F'' + \frac{m}{3} F''', \quad (25)$$

$$R_2 = \frac{m}{6} M_4 + \frac{m(m-1)}{8} (M_4 + 2M_5 + M_6). \quad (26)$$

The cubic derivative in (25) is included deliberately: the approximation is the full Taylor truncation through degree three. At order three,

$$A_3 = F - \frac{m}{2}F'' + \frac{m}{3}F''' + \left(\frac{m^2}{8} - \frac{m}{4}\right)F^{(4)} + \left(\frac{m}{5} - \frac{m^2}{6}\right)F^{(5)}, \quad (27)$$

$$B_{m,3} = \frac{m}{120} + \frac{5}{12}\binom{m}{2},$$

$$R_3 = B_{m,3}M_6 + \frac{\binom{m}{3}}{8}(M_6 + 3M_7 + 3M_8 + M_9). \quad (28)$$

For example, $\theta_{1,4} = 1/6$, $\theta_{1,6} = 1/120$, and $\theta_{2,6} = 5/12$ follow from (13). The supplied code computes these quantities as exact rationals rather than decimal approximations.

4.2 Comparison of the integrated bounds

The monotonicity (8) gives

$$R_s(m) \leq \left(B_{m,s} + \binom{m}{s}\right) M_{2s}(m) \leq \left(\sum_{j=1}^s \binom{m}{j}\right) M_{2s}(m). \quad (29)$$

Thus retaining the additional moments never worsens the corresponding coarse positive-moment bound. Direct integration of (4) would instead give $2s4^s m^s M_{2s}(m)$.

For completeness, the general finite-difference theorem of [9, Theorem 1.2, version 2] gives a remainder $33m^s \mathcal{E}_{2s}(m)$ for $m \geq 2$. In the measure setting,

$$\Delta^{2s} a_j = \int x^{2s}(1-x)^j d\mu(x), \quad \mathcal{E}_{2s}(m) = e^{-m} \sum_{j \geq 0} \frac{m^j}{j!} |\Delta^{2s} a_j| \leq M_{2s}^\mu(m).$$

Our radius is smaller than its computable unsigned-majorant specialization $33m^s M_{2s}(m)$. The original seminorm may exploit cancellation, so this is not a dominance claim against $33m^s \mathcal{E}_{2s}(m)$ itself.

5 The optimal scalar constant

5.1 Its leading term is best possible

Definition 5.1. The optimal scalar coefficient for the truncation at order s is

$$\mathcal{K}_{m,s} = \sup_{0 < x \leq 1} \frac{|E_m(x) - Q_{m,s}(x)|}{x^{2s}}. \quad (30)$$

The quotient has its removable value $|c_{2s}(m)|$ at $x = 0$.

Theorem 5.2 (Optimal leading coefficient). *For every fixed integer $s \geq 1$,*

$$\mathcal{K}_{m,s} \sim \frac{m^s}{2^s s!} \quad (m \rightarrow \infty). \quad (31)$$

Proof. Taking $x \downarrow 0$ in (30) and applying Lemma 3.1 gives the required lower limit.

For the upper limit, fix $\delta \in (0, 1)$. On $0 < x \leq \delta$, Theorem 3.2 gives

$$\frac{|E_m(x) - Q_{m,s}(x)|}{m^s x^{2s}} \leq \frac{B_{m,s}}{m^s} + \frac{\binom{m}{s}}{2^s m^s} (1 + \delta)^s.$$

Here $B_{m,s} = \mathcal{O}_s(m^{s-1})$ and $\binom{m}{s}/m^s \rightarrow 1/s!$. On $\delta \leq x \leq 1$, the kernel is between zero and one, while all retained coefficients have degree at most $s-1$ in m . Hence the supremum of the normalized quotient over that subinterval is $\mathcal{O}_{s,\delta}(m^{-1})$. Send m to infinity and then send δ to zero. $\square$

The supremum of the explicit polynomial envelope in (17) is not asymptotically the same as $\mathcal{K}_{m,s}$. The split in this proof keeps the additional information that the actual Taylor polynomial is small relative to m^s when x stays away from zero.

5.2 A second term and a unique worst argument

Theorem 5.3 (Sharp second-order asymptotics). *Fix $s \geq 1$. For all sufficiently large integers m , the quotient in (30) attains its global maximum at a unique point $x_{m,s} \in (0, 1)$. Moreover,*

$$x_{m,s} = \frac{2s(s+1)}{3m} + \mathcal{O}_s(m^{-2}), \quad (32)$$

$$\mathcal{K}_{m,s} = \frac{m^s}{2^s s!} \left(1 + \frac{s(-2s^2 + 5s + 1)}{9m} + \mathcal{O}_s(m^{-2}) \right). \quad (33)$$

More strongly, $mx_{m,s}$ and $m^{-s}\mathcal{K}_{m,s}$ are restrictions of real-analytic functions of $1/m$ near zero. Their expansions in $1/m$ therefore converge for sufficiently large m at each fixed s .

Proof. We first establish global localization, then use a local analytic maximum. Put $\varepsilon = m^{-1/2}$ and $y = x/\varepsilon$. In the physical range $0 \leq y < 1/\varepsilon$, set

$$f_\varepsilon(y) = \exp\left(\varepsilon^{-2}[\log(1 - \varepsilon y) + \varepsilon y]\right). \quad (34)$$

Its value at $y = 1/\varepsilon$ is zero. At $\varepsilon = 0$, define $f_0(y) = e^{-y^2/2}$. On every fixed compact interval, the series

$$\log f_\varepsilon(y) = -\frac{y^2}{2} - \sum_{k \geq 3} \frac{\varepsilon^{k-2} y^k}{k} \quad (35)$$

and any fixed number of its derivatives converge uniformly when $|\varepsilon y| \leq 1/2$.

Let $\mathbb{T}_{2s-1}$ denote Taylor truncation in y , and define the signed normalized remainder

$$H_\varepsilon(y) = (-1)^s \frac{f_\varepsilon(y) - \mathbb{T}_{2s-1}f_\varepsilon(y)}{y^{2s}}. \quad (36)$$

The value at $y = 0$ is removable. The integral Taylor-remainder formula shows that $H_\varepsilon \rightarrow H_0$ in C^2 on every fixed compact interval. Explicitly,

$$H_0(y) = \frac{1}{2^s(s-1)!} \int_0^1 (1-t)^{s-1} e^{-ty^2/2} dt. \quad (37)$$

This is positive and strictly decreasing for $y > 0$, with $H_0(0) = \kappa_s := 1/(2^s s!)$.

We also need control on intervals growing with m . Since $0 \leq f_\varepsilon(y) \leq e^{-y^2/2}$, and the scaled Taylor coefficients satisfy

$$c_{2j}(m)m^{-j} = \mathcal{O}_s(1), \quad c_{2j+1}(m)m^{-j-1/2} = \mathcal{O}_s(\varepsilon),$$

there is a constant C_s such that for $R \geq 1$ and $R \leq y \leq 1/\varepsilon$,

$$|H_\varepsilon(y)| \leq \frac{C_s}{R^2} + \frac{C_s \varepsilon}{R}. \quad (38)$$

For the even terms the largest retained degree is $2s - 2$; for the odd terms it is $2s - 1$. These two facts give the two powers of R in (38). The constant term and the kernel are covered by the same estimate.

Fix a small $\delta > 0$. Choose a large fixed R so that the tail bound is smaller than $\kappa_s/2$. On $[\delta, R]$, the strict decrease in (37) and uniform convergence give a fixed gap below κ_s . Since $H_\varepsilon(0) \rightarrow \kappa_s$, every global maximizer of $|H_\varepsilon|$ eventually lies in $[0, \delta]$.

The series (35) is jointly analytic near $(\varepsilon, y) = (0, 0)$. Removing the Taylor terms and dividing by y^{2s} preserves that analyticity. There is also the symmetry

$$H_{-\varepsilon}(-y) = H_\varepsilon(y). \quad (39)$$

The first terms of the local expansion are

$$H_\varepsilon(y) = \kappa_s - D_s \varepsilon^2 + U_s \varepsilon y - V_s y^2 + \mathcal{O}_s((|\varepsilon| + |y|)^4), \quad (40)$$

where

$$\begin{aligned} U_s &= \frac{1}{3 \cdot 2^{s-1} (s-1)!}, & V_s &= \frac{1}{2^{s+1} (s+1)!}, \\ D_s &= \kappa_s \frac{s(s-1)(4s+1)}{9}. \end{aligned}$$

To obtain D_s , the next coefficient of c_{2s} has either one part of size four and $s - 2$ parts of size two, or two parts of size three and $s - 3$ parts of size two in the exponential in Lemma 3.1. An impossible part pattern contributes zero. The coefficient U_s comes from one part of size three in c_{2s+1} ; $-V_s$ comes from the leading term of c_{2s+2} . The symmetry (39) rules out odd total degree in (40).

Since $H_0''(0) = -2V_s < 0$, take δ small enough that H_0 is strictly concave and positive on $[0, \delta]$. The C^2 convergence makes the same true for H_ε at small positive ε . Moreover, $H_\varepsilon'(0) = U_s \varepsilon + \mathcal{O}_s(\varepsilon^3) > 0$ and $H_\varepsilon'(\delta) < 0$. There is a unique maximum in the interval, and global localization makes it the unique global maximum of the absolute quotient.

The analytic implicit-function theorem applied to $\partial_y H_\varepsilon(y) = 0$ gives an analytic critical point $y(\varepsilon)$ near zero. By symmetry and uniqueness it is odd. Therefore

$$y(\varepsilon) = \frac{U_s}{2V_s} \varepsilon + \mathcal{O}_s(\varepsilon^3) = \frac{2s(s+1)}{3} \varepsilon + \mathcal{O}_s(\varepsilon^3).$$

Substituting in (40) gives

$$H_\varepsilon(y(\varepsilon)) = \kappa_s + \left(\frac{U_s^2}{4V_s} - D_s \right) \varepsilon^2 + \mathcal{O}_s(\varepsilon^4).$$

The quotient of the parenthesized coefficient by κ_s is $s(-2s^2 + 5s + 1)/9$. Finally, $x_{m,s} = \varepsilon y(\varepsilon)$ and $\mathcal{K}_{m,s}/m^s = H_\varepsilon(y(\varepsilon))$. The former divided by ε^2 , and the latter, are even analytic functions of ε . They are analytic in $\varepsilon^2 = 1/m$, which proves the stronger assertion. $\square$

For the first three orders, the optimal constants are

$$\begin{aligned} \mathcal{K}_{m,1} &= \frac{m}{2} + \frac{2}{9} + \mathcal{O}(m^{-1}), \\ \mathcal{K}_{m,2} &= \frac{m^2}{8} + \frac{m}{12} + \mathcal{O}(1), \\ \mathcal{K}_{m,3} &= \frac{m^3}{48} - \frac{m^2}{72} + \mathcal{O}(m). \end{aligned}$$

The negative second term at $s = 3$ is consistent with an optimal positive constant: the leading cubic term dominates. Neither this expansion nor its uniqueness threshold is asserted uniformly as s increases.

5.3 A simple finite improvement at order one

At the first order one can also obtain a useful closed upper bound for every m , without an unspecified asymptotic threshold.

Proposition 5.4 (A finite coefficient for the Poisson product). *For every integer $m \geq 1$,*

$$\mathcal{K}_{m,1} \leq \frac{m^2}{2m-1}. \quad (41)$$

For $m \geq 2$ the maximizing argument is unique; for $m = 1$ it is $x = 1$. Consequently,

$$|P_m - F(m)| \leq \min \left\{ \frac{m}{2}(M_2 + M_3), \frac{m^2}{2m-1}M_2 \right\}. \quad (42)$$

Proof. On $0 < x < 1$, put $r(x) = (1 - E_m(x))/x^2$ and $D(x) = mx^2 - 2x + 2$. A direct differentiation shows that the sign of $r'(x)$ is the sign of

$$\ell(x) = mx + (m-1)\log(1-x) + \log(D(x)/2).$$

In fact, $x^3 r'(x) = 2(e^{\ell(x)} - 1)$. Another differentiation gives

$$\ell'(x) = \frac{mx^2(1-mx)}{(1-x)D(x)}.$$

For $m \geq 2$, the function ℓ starts at zero, increases strictly to $x = 1/m$, and then decreases strictly to $-\infty$. Thus it has exactly one zero x_* in $(1/m, 1)$, giving the unique maximum of r . At that point $E_m(x_*) = 2(1 - x_*)/D(x_*)$, and hence $r(x_*) = m/D(x_*)$. Completing the square gives $D(x) \geq 2 - 1/m$, which proves (41). When $m = 1$, the displayed derivative is positive on $(0, 1)$ and $r(1) = 1$, giving the same bound. Multiplication by e^{-mx} and signed integration prove the second term in (42); the first is (24). $\square$

The numerical tables use the unified radius R_s from (22) at every order. Formula (42) provides an additional, inexpensive refinement for a calculation using the Poisson product alone.

For a particularly simple implementation, write $q_i = e^{-mp_i}$ and $Z = \prod_i (1 + q_i)$. Expansion of the second moment gives

$$M_2(m) = Z \left[\left(\sum_i \frac{p_i q_i}{1 + q_i} \right)^2 + \sum_i \frac{p_i^2 q_i}{(1 + q_i)^2} \right]. \quad (43)$$

Thus the second bound in (42) can be evaluated using three scalar accumulations after the exponentials, without even constructing a higher-order jet.

6 Uniform control for arbitrary probability vectors

An error formula becomes more informative when its scale is explicit. The natural parameter here is

$$\lambda = \lambda(m, p) = \sum_{i=1}^n e^{-mp_i}, \quad (44)$$

the expected number of missing coupons in the Poissonized model. We consider $\lambda \leq L$ with L fixed. This includes the usual coverage window and every later time. No restriction is imposed on the ratio of the largest and smallest probabilities.

Let

$$\mathcal{T}_k(L) = \sum_{j=1}^k \left\{ \begin{matrix} k \\ j \end{matrix} \right\} L^j, \quad k \geq 1, \quad (45)$$

where the braces denote Stirling numbers of the second kind. These Touchard polynomials arise naturally when expanding moments of sums of independent indicators.

Lemma 6.1 (A uniform moment envelope). *Suppose $\lambda \leq L < n$, and set $h = \log(n/L) > 0$. For every integer $k \geq 1$,*

$$M_k(m) \leq \frac{W_k(L, h)}{m^k}, \quad W_k(L, h) := e^L \mathcal{T}_k(L)(h+k)^k. \quad (46)$$

Also $m \geq nh$.

Proof. Write $q_i = e^{-mp_i}$ and $u_i = mp_i = -\log q_i$. Put $Z = \prod_i (1+q_i) \leq e^\lambda \leq e^L$ and introduce independent Bernoulli variables Y_i with probabilities $q_i/(1+q_i)$. For $U = \sum_i u_i Y_i$, the product expansion gives

$$M_k(m) = Z m^{-k} \mathbb{E} U^k. \quad (47)$$

The weighted tail satisfies

$$\sum_i q_i \mathbf{1}_{\{u_i \geq t\}} \leq \min(L, n e^{-t}), \quad t \geq 0. \quad (48)$$

For $j \geq 1$, integration of this tail yields

$$\begin{aligned} a_j &:= \sum_i q_i u_i^j \leq L h^j + j n \int_h^\infty t^{j-1} e^{-t} dt \\ &= L \mathbb{E}(h+V)^j \leq L(h+j)^j, \end{aligned} \quad (49)$$

where V is exponential with mean one. The final inequality follows by expanding the moment and using $a! \leq j^a$ for $0 \leq a \leq j$.

Expand U^k and group equal indices by a set partition of $[k]$. In each block assignment, dropping the restriction that different blocks receive distinct indices can only increase the nonnegative sum. Since $q_i/(1+q_i) \leq q_i$, a partition with b blocks contributes at most $L^b(h+k)^k$ by (49). Summing over partitions proves

$$\mathbb{E} U^k \leq \mathcal{T}_k(L)(h+k)^k.$$

Combining this with (47) proves the moment envelope. Finally, convexity of the exponential gives $\lambda \geq n e^{-m/n}$, so $m \geq n \log(n/L)$. $\square$

Theorem 6.2 (An explicit uniform coverage-window bound). *Fix $s \geq 1$ and $L > 0$. Define*

$$b_s = \sum_{j=1}^{s-1} \frac{\theta_{j,2s}}{j!},$$

with $b_1 = 0$. If $n > L$ and $\lambda(m, p) \leq L$, then

$$\begin{aligned} |P_m - A_s(m)| &\leq R_s(m) \leq \frac{b_s W_{2s}(L, h)}{(nh)^{s+1}} \\ &\quad + \frac{1}{2^s s!} \sum_{\ell=0}^s \binom{s}{\ell} \frac{W_{2s+\ell}(L, h)}{(nh)^{s+\ell}}, \quad h = \log(n/L). \end{aligned} \quad (50)$$

Consequently,

$$\sup_{\substack{p_i \geq 0, \sum_i p_i = 1 \\ m \in \mathbb{N}, \lambda(m, p) \leq L}} |P_m - A_s(m)| = \mathcal{O}_{L,s} \left(\left(\frac{\log n}{n} \right)^s \right). \quad (51)$$

More precisely, the right-hand side of (50) is

$$\frac{e^L \mathcal{T}_{2s}(L)}{2^s s!} \left(\frac{h}{n} \right)^s \left(1 + \mathcal{O}_{L,s}(h^{-1}) \right). \quad (52)$$

Proof. For $m \geq 1$, we have $B_{m,s} \leq b_s m^{s-1}$ and $\binom{m}{s} \leq m^s/s!$. Insert (46) into (22), and then use $m \geq nh$ in every denominator. This proves (50). At fixed s, L , its $\ell = 0$ term has the leading expression (52). The $\ell \geq 1$ terms lose at least one power of n , while the b_s term loses an additional factor nh . Expanding $(h + 2s)^{2s}$ gives the stated relative $\mathcal{O}(h^{-1})$ term. $\square$

The finite inequality (50), rather than its asymptotic abbreviation, is the explicit guarantee. Its leading coefficient is an upper envelope for the certificate. We do not claim that $e^L \mathcal{T}_{2s}(L)/(2^s s!)$ is the best constant for the signed probability error.

The condition $\lambda \leq L$ is inexpensive to check. It includes probability vectors with a few exceptionally rare coupons, many common coupons, and no uniform lower bound on individual probabilities. The result also permits m to be arbitrarily large; the proof uses a fixed L rather than replacing it by the potentially very small actual λ inside a logarithm.

7 Sharpness within the coupon model

There are two distinct sharpness questions. The first concerns the leading constant of the positive moment certificate. The second concerns the power of n in the coverage-window rate for the specified truncation. Different examples answer them.

7.1 A rare coupon attains the certificate

Theorem 7.1 (Asymptotically exact positive radius). *Fix $a > 0$ and an integer $s \geq 1$. For integers $m > a$, take two coupons with probabilities*

$$p_1 = \frac{a}{m}, \quad p_2 = 1 - \frac{a}{m}.$$

Then

$$P_m - A_s(m) \sim \frac{(-1)^{s+1} e^{-a} a^{2s}}{2^s s! m^s}, \quad (53)$$

$$R_s(m) \sim \frac{e^{-a} a^{2s}}{2^s s! m^s}. \quad (54)$$

In particular, $|P_m - A_s(m)|/R_s(m) \rightarrow 1$.

Proof. The four subset probabilities are 0, a/m , $1 - a/m$, and 1. For every fixed $k \geq 1$,

$$M_k(m) = e^{-a} (a/m)^k + \mathcal{O}_{k,a}(e^{-m+a}),$$

and

$$F^{(2s)}(m) = -e^{-a} (a/m)^{2s} + \mathcal{O}_{s,a}(e^{-m+a}).$$

Compare the two consecutive approximants:

$$P_m - A_s(m) = c_{2s}(m) F^{(2s)}(m) - c_{2s+1}(m) F^{(2s+1)}(m) + \mathcal{O}(R_{s+1}(m)). \quad (55)$$

Here $c_{2s+1}(m) = \mathcal{O}_s(m^s)$, and the moment formula shows $R_{s+1}(m) = \mathcal{O}_{s,a}(m^{-s-1})$. Using (12) in (55) proves (53). In (22), the term $2^{-s} \binom{m}{s} M_{2s}$ dominates: the $B_{m,s}$ term is smaller by a factor $\mathcal{O}(1/m)$, as are all terms with $\ell \geq 1$. This proves (54). $\square$

This example is a sharpness construction, not a restriction on the algorithm. The input distribution varies with m , as is appropriate for a bound uniform over all input vectors. It shows that the leading coefficient cannot be reduced even if the abstract signed-measure setting is restricted to actual coupon probabilities.

7.2 The uniform rate has the correct order

Theorem 7.2 (The first omitted term in the usual window). *Let $p_i = 1/n$, let $s \geq 1$ be fixed, and set*

$$m_n = \lfloor n(\log n + c) \rfloor, \quad g(c) = \exp(-e^{-c}).$$

Then

$$\frac{P_{m_n} - A_s(m_n)}{(\log n/n)^s} \longrightarrow \frac{(-1)^s}{2^s s!} g^{(2s)}(c). \quad (56)$$

At the single choice $c = \log 2$, this limit is nonzero for every $s \geq 1$.

Proof. Put $g_n(z) = (1 - e^{-z}/n)^n$. Since $F_n(t) = g_n(t/n - \log n)$, writing $c_n = m_n/n - \log n \rightarrow c$ gives

$$F_n^{(k)}(m_n) = n^{-k} g_n^{(k)}(c_n).$$

For each fixed k , these last derivatives converge to $g^{(k)}(c)$. For example, the identity $n \log(1 - u/n) = -u + \mathcal{O}(n^{-1})$ with all fixed derivatives uniform on compact u -intervals proves this after setting $u = e^{-z}$.

Apply (55) at $m = m_n$. The leading even coefficient contributes the right-hand side of (56), because $m_n \sim n \log n$. The next, odd derivative term is smaller by a factor $\mathcal{O}(1/n)$ after normalization. The remainder from Theorem 6.2 is smaller by a factor $\mathcal{O}(\log n/n)$. This proves the limit.

For the nonvanishing assertion, repeated differentiation gives

$$g^{(2s)}(c) = e^{-u} \mathcal{T}_{2s}(-u), \quad u = e^{-c}.$$

The polynomial $\mathcal{T}_{2s}$ is monic with integer coefficients. By the rational-root theorem it cannot vanish at the nonintegral rational number $-1/2$. Taking $c = \log 2$ completes the proof. $\square$

For any prescribed $L > 0$, the same argument gives a nonzero lower example inside $\lambda \leq L$: choose an integer $q \geq 2$ with $1/q < L$ and set $c = \log q$. Then $\lambda(m_n, p) \rightarrow 1/q < L$ and $\mathcal{T}_{2s}(-1/q) \neq 0$. Thus the power in (51) is sharp for each fixed positive L .

At some other locations the displayed derivative does vanish. In particular, $g''(0) = 0$, so an experiment centered only at $c = 0$ can misrepresent the generic leading error at order one. This is why the uniform numerical experiments use $c = \log 2$.

The lower example concerns the specified Poisson–Charlier truncation. It is not a lower bound on all algorithms for coupon collection. The classical expansions already explain the existence of such fixed-order rates; the explicit finite radius is the quantitative refinement studied here.

8 Computation in linear work at fixed order

8.1 Products replace exponentially many subsets

The definitions of $F^{(k)}$ and M_k appear to require 2^n subsets. Their generating products avoid this enumeration. Put $u_i = mp_i$ and $q_i = e^{-u_i}$, and form the dimensionless series

$$\mathcal{F}(z) = \prod_{i=1}^n (1 - q_i e^{u_i z}) = F(m - mz), \quad (57)$$

$$\mathcal{Z}(z) = \prod_{i=1}^n (1 + q_i e^{u_i z}). \quad (58)$$

If $f_k = [z^k] \mathcal{F}(z)$ and $z_k = [z^k] \mathcal{Z}(z)$, then

$$f_k = \frac{(-1)^k m^k}{k!} F^{(k)}(m), \quad z_k = \frac{m^k}{k!} M_k(m). \quad (59)$$

Thus

$$A_s(m) = \sum_{k=0}^{2s-1} \frac{k! c_k(m)}{m^k} f_k. \quad (60)$$

The signed product is needed only through degree $2s - 1$, and the positive product only through degree $3s$. Higher coefficients cannot influence lower ones, so truncating after every multiplication is exact at the level of formal power series.

For one input probability, compute the series coefficients of $q_i e^{u_i z}$ by $v_0 = q_i$, $v_k = v_{k-1} u_i / k$. The factor for $\mathcal{F}$ is $(1 - q_i, -v_1, -v_2, \dots)$ and the factor for $\mathcal{Z}$ is $(1 + q_i, v_1, v_2, \dots)$. The second product has only nonnegative coefficients, which is especially convenient for a positive remainder radius.

Proposition 8.1 (Arithmetic cost). *With naive truncated convolution, the two product calculations require $\mathcal{O}(ns^2)$ arithmetic operations and n exponential evaluations. The Charlier coefficients require $\mathcal{O}(s)$ further operations. The supplied exact-rational construction of all $\theta_{j,2s}$ costs $\mathcal{O}(s^3)$ operations and can be cached by order. Consequently, the total arithmetic work is*

$$\mathcal{O}(ns^2 + s^3), \quad (61)$$

with $\mathcal{O}(s)$ coefficient working memory beyond the input and cache. For fixed s , the work is linear in the number of probabilities.

Proof. A truncated convolution of degree d has at most $(d + 1)^2$ scalar products. Here $d \leq 3s$. Update the two products once per input. The coefficient recurrence (11) uses a running prefix sum. To construct the tails, successively multiply $1, d, d^2, \dots, d^{s-1}$ truncated at degree $2s - 1$ and retain the sum of each coefficient list. This has $s - 1$ convolutions of quadratic cost. Each current coefficient list has length $\mathcal{O}(s)$. $\square$

Equal probabilities can be grouped. Raising one truncated factor to its multiplicity by repeated squaring often makes the uniform case very fast. The timing experiment disables this optimization to test the general pass over the input.

8.2 Rounding belongs in the certificate

The analytic inequality alone does not certify a floating-point result. The implementation in `src/coupon_certificate.py` therefore uses Arb real balls through python-flint [6, 5]. Integer or rational input weights are normalized exactly. The conversion to real balls, all exponentials,

and all series operations return enclosures. If $\mathbf{A}$ encloses A_s and $\mathbf{R}$ encloses the nonnegative radius R_s , the program returns

$$\mathbf{A} + [-\text{upper}(\mathbf{R}), \text{upper}(\mathbf{R})]. \quad (62)$$

The interval can then be intersected with $[0, 1]$.

The program rejects binary floating-point weights to avoid silently reinterpreting rounded input as exact data. For example, a user who intends the exact decimal 0.1 supplies the string "0.1" or a rational such as "1/10". The output is a real-ball string; plain floating-point midpoints in the plotting data are labeled as such and are not numerical certificates.

Increasing the working precision reduces numerical uncertainty. It does not shrink the analytic remainder at a fixed order. Increasing the order can reduce that remainder in the asymptotic region, but a finite input is not promised a monotone improvement at every successive order. The code returns the actual computed radius so that either limitation is visible.

8.3 What the algorithmic statement means

For fixed s and fixed L , the calculation gives certified absolute accuracy on the scale $(\log n/n)^s$ in the bounded- λ region, with one pass over the categories. The work does not contain a factor proportional to the number of observations m . A large m still affects the precision and cost of the underlying arithmetic, so (61) does not mean constant cost for arbitrarily long binary integers.

This is a deterministic approximation with a verified radius, not an exact counting algorithm for all nonuniform instances. Other coupon questions, including expected collection times and general multiplicities, have their own approximation and computational literature [1, 2]. The claimed improvement is the remainder guarantee for the fixed-time probability calculation.

9 Reproducible computations

9.1 Validation independent of the approximation

The package separates three forms of evidence.

1. **Exact small instances.** The test suite uses integer inclusion–exclusion and independent occupancy-state recurrences. It tests unequal rational probabilities, impossible coverage times, zero probabilities, $n = 1$, $m < s$, and orders one through five. Grouped and ungrouped products are checked against direct subset calculations.
2. **High-precision probability references.** The experiment harness evaluates full grouped inclusion–exclusion at 320-bit precision. All 68 reference intervals were wholly contained in the corresponding 224-bit certificates: 24 uniform instances, 20 rare-coupon instances, and 24 heterogeneous instances.
3. **Scalar asymptotics.** A separate SymPy and mpmath program checks exact coefficient identities for $s = 1, \dots, 6$ and performs 12 high-precision stationary-point checks. These computations illustrate Theorem 5.3; its global assertion is proved analytically.

The standard test command completed 11 test methods successfully. Some methods contain loops over many inputs; the number of methods is not the number of evaluated instances. Computations support the implementation and expose mistakes; they are not substitutes for the proofs above.

9.2 Uniform coupons

For each $n \in \{50, 100, 300, 1000, 3000, 10000\}$, the experiment uses $m = \lfloor n(\log n + \log 2) \rfloor$ and orders one through four. Symmetry reduces the independent reference to the complete $n + 1$ term sum

$$P_m = \sum_{j=0}^n (-1)^j \binom{n}{j} (1 - j/n)^m. \quad (63)$$

Every term is evaluated; this reference does not use the moment certificate.

Order s	Absolute error	Radius R_s	Prior majorant / R_s
1	$7.512657 \cdot 10^{-5}$	$6.124396 \cdot 10^{-4}$	1.9996
2	$4.184661 \cdot 10^{-8}$	$6.193232 \cdot 10^{-7}$	511.73
3	$4.787005 \cdot 10^{-12}$	$8.030013 \cdot 10^{-10}$	18 412.21
4	$1.833985 \cdot 10^{-14}$	$1.226568 \cdot 10^{-12}$	784 901.52

Table 1: Uniform collection with $n = 10\,000$ and $m = 99\,034$. Errors and radii are rounded for display; complete real-ball values are in the data files. The prior comparison uses mM_2 at order one (HLZ Lemma 2.1) and $2s4^s m^s M_{2s}$ at higher orders (HLZ Lemma 5.3).

The radius can be much larger than the actual error because the positive moment sum discards cancellation. At order four in Table 1, it still certifies absolute error below $1.23 \cdot 10^{-12}$ while the generic earlier scalar majorant is about $7.85 \cdot 10^5$ times larger. This is a comparison of proved upper bounds, not a claim that the numerical approximation itself improved by that factor.

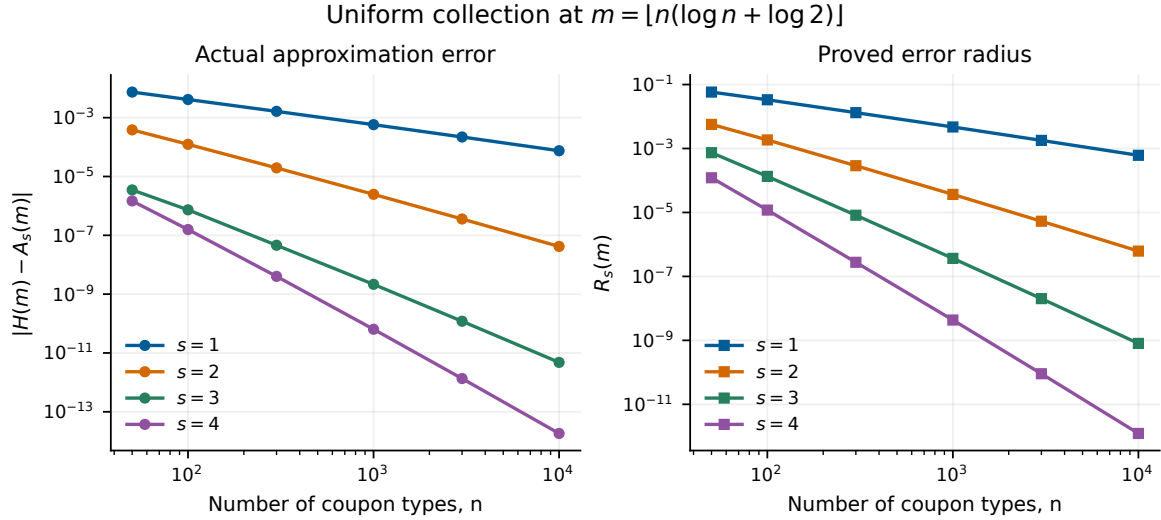


Figure 1: Uniform collection in the fixed window $c = \log 2$. The left panel shows the absolute approximation errors; the right panel shows their certified radii. Points come from the recorded experiments, with full grouped inclusion–exclusion references. The distinct vertical scales make the conservatism of a positive radius visible.

9.3 Sharpness and unequal probabilities

The rare-coupon experiment uses $p_1 = 1/m$, $p_2 = 1 - 1/m$ for $m = 10^2, \dots, 10^6$. At $m = 10^6$, the ratios $|P_m - A_s|/R_s$ for $s = 1, 2, 3, 4$ are approximately

$$0.999999416667, \quad 0.999996833341, \quad 0.999983208517, \quad 0.999941902368.$$

These approach one as asserted in Theorem 7.1. The smallest radius in these four cases is about $9.58 \cdot 10^{-28}$; the reference and certificate precisions resolve it.

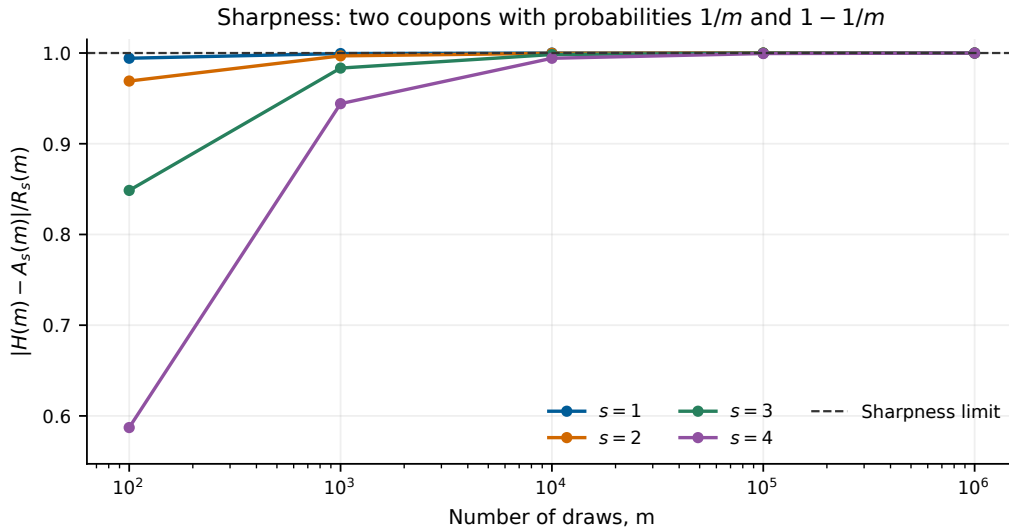


Figure 2: Asymptotic attainment of the positive radius in the rare-coupon family. The horizontal line marks ratio one. Each reference is a complete four-term inclusion–exclusion calculation. The limiting assertion is Theorem 7.1.

Two further 100-coupon models test unequal probabilities. The first has 50 coupons of weight one and 50 of weight three. The second has one coupon of weight one and 99 coupons of weight 100. The weights are normalized exactly. For each model, three integer times are chosen by target Poissonized missing means 2, 0.5, and 0.05, and all four orders are evaluated. The references contain 2601 and 200 grouped terms, respectively. All reference intervals satisfy containment. The recorded data retain the exact inputs, achieved missing means, error intervals, and bound comparisons; the target missing mean is not substituted for the achieved value in any certificate.

9.4 Running time and numerical environment

Timing inputs have $w_i = 1 + (i \bmod 997)$ for $i = 0, \dots, n - 1$. The integer m is chosen so that $\lambda \leq 0.5$. Equal-weight grouping is disabled, and timing includes exact normalization and certificate construction but excludes the selection of m . The implementation used Python 3.12.14, python-flint 0.9.0, one FLINT thread, and 128-bit arithmetic.

Number of coupons	Order 2 time (s)	Order 3 time (s)
100	0.0036	0.0049
1000	0.0295	0.0459
10000	0.1991	0.3450
100000	2.3039	4.0710

Table 2: One measured run per point after a warm-up. These observations illustrate the arithmetic scaling; they are not statistical performance estimates or bit-complexity bounds.

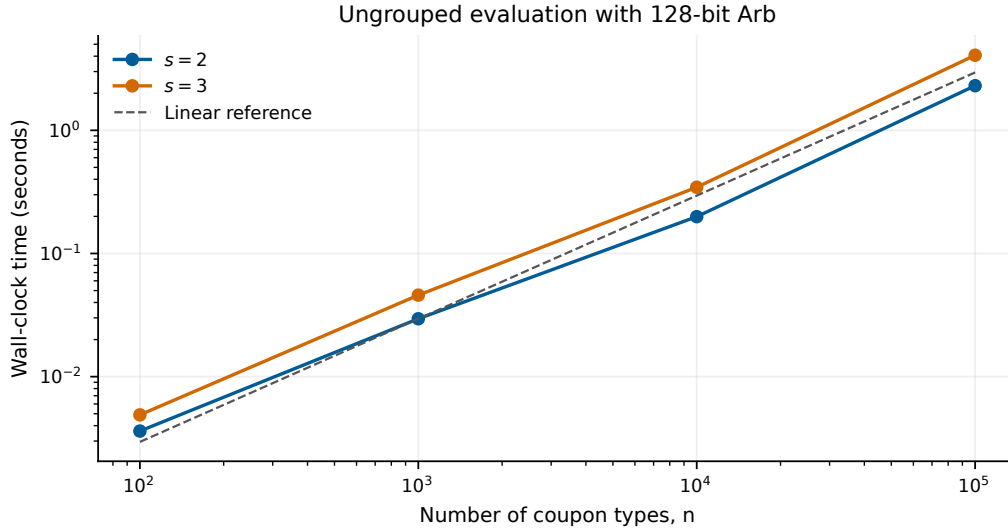


Figure 3: Measured ungrouped evaluation times. The dashed line is proportional to n and is anchored at the order-two, $n = 1000$ measurement. The proof of Proposition 8.1 is independent of the timing experiment.

10 Consequences and practical interpretation

10.1 From probabilities to sample-size decisions

A certified probability interval can answer a threshold question: has the chance of complete coverage reached a chosen target α ? If the lower endpoint is at least α , the answer is yes. If the upper endpoint is below α , the answer is no. An interval containing α leaves that particular query unresolved and calls for a higher order, more arithmetic precision when rounding is relevant, or a different bound.

Monotonicity of P_m makes it possible to maintain a certified bracket for the smallest qualifying integer m . A binary-search implementation must preserve that bracket and must not decide from the midpoint of A_s alone. The approximation $A_s(m)$ itself is not asserted to be monotone, and no universal exact-quantile complexity follows from this observation. In particular, a threshold exactly equal to a probability requires an equality-aware treatment.

10.2 Why more moments improve a bound

The strongest part of (22) is

$$\frac{\binom{m}{s}}{2^s} (M_{2s} + sM_{2s+1} + \cdots + M_{3s}).$$

Replacing every higher moment by M_{2s} throws away the information that relevant subset probabilities can be small. When those subsets are small, $M_{2s+\ell}/M_{2s}$ is small for $\ell > 0$, retaining the factor 2^{-s} . The rare-coupon family makes this observation exact. When subset probabilities are not small, the same finite formula remains valid and records the larger uncertainty.

The mechanism is therefore quantitative rather than heuristic: positivity of the coefficients gives a global remainder inequality, and the input moments adapt it to the particular probability vector.

10.3 An all-orders scalar calculation

The analytic statement in Theorem 5.3 also gives a procedure for calculating more terms of the optimal constant. Expand $H_\varepsilon(y)$ from (35); substitute an odd series $y(\varepsilon) = b_1\varepsilon + b_3\varepsilon^3 + \dots$; solve $\partial_y H_\varepsilon = 0$ successively; and substitute back. The nonzero coefficient $-2V_s$ makes every step linear in the next unknown coefficient. All inputs are rational at fixed s , so every coefficient obtained is rational.

This convergent local expansion describes the exact global optimum for sufficiently large m . A finite truncation of that expansion is not automatically an upper bound: it needs its own explicit remainder before being used as a numerical certificate. The delivered probability implementation uses the all- m inequality (22).

11 Further research questions

The following questions grow directly from proved statements in this paper. They have different levels of difficulty and should remain separate from established results.

- Q1. Explicit finite- m optimal constants.** Can one bound $\mathcal{K}_{m,s}$ tightly with a small number of operations, including the range before the asymptotic unique maximum is guaranteed? A certified one-dimensional optimization could be useful, but an analytic formula with a verified remainder would be more reusable.
- Q2. Growing correction order.** The proofs keep s fixed. Determine a quantitative range $s = s(m)$ in which the peak localization, analytic coefficient estimates, and moment certificate remain effective. This is the step needed for a meaningful precision-versus-work theorem when the requested number of digits grows with the input size.
- Q3. Cancellation without exponential enumeration.** The unsigned product loses cancellations between subsets of opposite parity. Can additional efficiently computable signed information yield a substantially smaller positive radius, especially for uniform or nearly uniform inputs? Any such refinement should preserve a complete error proof.
- Q4. Relative error for small coverage probabilities.** The present radius is additive. Find conditions under which it is at most εP_m , or construct a separate factorized certificate that remains useful before the bounded- λ region. Highly unequal probabilities are the essential test.
- Q5. Optimal constants over probability vectors.** For fixed L, s , determine the asymptotic supremum of $|P_m - A_s|/(\log n/n)^s$ subject to $\lambda \leq L$. The Touchard envelope proves an upper bound, and the uniform family gives nonzero lower examples, but these constants have not been matched.
- Q6. Heterogeneous collection quotas.** Requiring several copies of each category replaces each Poisson factor by a Poisson upper-tail probability. Which parts of the positive coefficient argument survive, and what extra data suffice to certify a fixed-sample probability efficiently?
- Q7. Bit complexity and automatic precision.** Establish the arithmetic precision needed to make numerical interval width comparable to the analytic radius for rational inputs of bounded encoding length. This would convert the present arithmetic count into a rigorous bit-complexity statement.
- Q8. Certified quantiles and sensitivity.** Combine the intervals with monotonicity and derivative information to bound coverage-time quantiles and their sensitivity to errors in the input probabilities. Exact threshold ties and uncertain input probabilities require explicit treatment.

Q9. Formal verification. The finite scalar theorem has a short algebraic structure: coefficient positivity, a binomial remainder, and a finite signed sum. It is a plausible first target for a Lean formalization. The analytic maximum theorem would require a substantially richer formal development.

A Reproduction and file guide

The package contains the LaTeX manuscript and bibliography, the compiled PDF, the core implementation, an independent test suite, two experiment programs, recorded data, and the plotting assets. No external manuscript is bundled as if it were authored here.

From the package directory, install the dependencies and run:

```
python -m pip install -r requirements.txt
python -m unittest discover -s tests -v
python experiments/run_experiments.py
python experiments/sharp_scalar.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -cd article/coupon_collector.tex
```

To calculate a single certificate:

```
python src/coupon_certificate.py --uniform 10000 \
  --m 99034 --order 4 --bits 224
python src/coupon_certificate.py --weights 1,2,3,5 \
  --m 100 --order 3 --bits 160
```

The experiments record the exact input descriptions, software versions, working precision, reference method, and a source hash. Timing data are machine-specific. Re-running the scripts overwrites their generated data and figures with the new run's measurements.

Path	Purpose
article/coupon_collector.tex	Complete manuscript source.
article/references.bib	Primary references and version information.
src/coupon_certificate.py	Rational inputs, product jets, and Arb enclosures.
tests/test_certificate.py	Exact and independent reference checks.
experiments/run_experiments.py	Probability experiments, running times, and figures.
experiments/sharp_scalar.py	Symbolic coefficients and scalar stationary-point checks.
results/experiments.json	Full probability and timing records, including ball strings.
results/*.csv	Tabular observations for reuse.
figures/	PDF and PNG versions of all three figures.

B A compact checklist of theorem domains

Result	Assumptions and meaning
Theorems 3.2 and 4.1	Positive integers m, s ; all $x \in [0, 1]$ and all probability vectors. Finite, explicit absolute-error inequalities.
Theorems 5.2 and 5.3	s fixed, $m \rightarrow \infty$ through integers. The global maximizer is unique only for sufficiently large m . The analytic radius in $1/m$ depends on s .
Theorem 6.2	s, L fixed; $n > L$; $\sum_i e^{-mp_i} \leq L$. The finite bound is explicit, uniform in p and in every qualifying m .
Theorem 7.1	A distribution varying with m . The ratio error/radius tends to one for this family.
Theorem 7.2	Uniform probabilities and fixed c . Sharpness concerns the chosen truncation, not every possible algorithm.
Proposition 8.1	Arithmetic operations and exponential evaluations. Extra bit costs and precision requirements are not suppressed as a theorem.

The directly strengthened statement is the scalar remainder lemma. The probability certificate, optimal-constant analysis, sharpness constructions, and numerical implementation make that improvement concrete and independently assessable.

References

- [1] Arnon Boneh and Micha Hofri. The coupon-collector problem revisited—a survey of engineering problems and computational methods. *Communications in Statistics. Stochastic Models*, 13(1):39–66, 1997. doi: 10.1080/15326349708807412. URL <https://web.cs.wpi.edu/~hofri/CCP.pdf>. Author-hosted revised manuscript. Accessed 8 October 2026.
- [2] Aristides V. Doumas and Vassilis G. Papanicolaou. The coupon collector’s problem revisited: generalizing the double Dixie cup problem of Newman and Shepp. *ESAIM: Probability and Statistics*, 20:367–399, 2016. doi: 10.1051/ps/2016016. URL <https://www.numdam.org/articles/10.1051/ps/2016016/>. Preprint first posted 11 December 2014. Accessed 8 October 2026.
- [3] Hsien-Kuei Hwang, Chong-Yi Li, and Vytas Zacharovas. Elementary asymptotics for the Stirling numbers of the second kind: The central range, 2026. URL <https://arxiv.org/abs/2605.29633v1>. Version 1, 28 May 2026. Lemma 5.3, equation (46). Accessed 8 October 2026.
- [4] Philippe Jacquet and Wojciech Szpankowski. Analytical depoissonization and its applications. *Theoretical Computer Science*, 201(1–2):1–62, 1998. doi: 10.1016/S0304-3975(97)00167-9. URL [https://doi.org/10.1016/S0304-3975\(97\)00167-9](https://doi.org/10.1016/S0304-3975(97)00167-9). Accessed 8 October 2026.
- [5] Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633. URL <https://arxiv.org/abs/1611.02831>. Accessed 8 October 2026.
- [6] Fredrik Johansson and the python-flint contributors. *python-flint 0.9.0 documentation*. The python-flint project, 2026. URL <https://python-flint.readthedocs.io/en/latest/>. Version 0.9.0, as displayed and used for the accompanying computations. Accessed 8 October 2026.

- [7] OpenAI. openai/math: Mathematical manuscripts and supporting proof artifacts. GitHub repository, 2026. URL <https://github.com/openai/math/tree/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb>. Commit fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb. Accessed 8 October 2026.
- [8] Anna Pósfai. *Approximation Theorems Related to the Coupon Collector's Problem*. PhD thesis, University of Szeged, 2010. URL <https://arxiv.org/abs/1006.3531>. Accessed 8 October 2026.
- [9] Vytas Zacharovas. An elementary approach to depoissonization, 2026. URL <https://arxiv.org/abs/2511.14324v2>. Version 2, 3 September 2026; first posted 18 November 2025. Theorem 1.2 uses the constant 33. Accessed 8 October 2026.